



Cultural analytics amid the rise of generative AI: critical insights for human-AI cocreative cultural studies

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Abstract

This essay reviews *Cultural Analytics* by Lev Manovich as a foundational text that not only charts the historical and technical emergence of cultural analytics but also provokes a deeper interrogation of how digital and generative AI-driven methods reconfigure the very ontology, cognition, and ethics of cultural inquiry. Cultural analytics combines computer science, data visualization, and media arts to study cultural phenomena at scale; generative AI further extends this paradigm by producing new artifacts that blur boundaries of authorship and challenge traditional categories. This review, following Manovich’s admonition throughout the book, concludes that that meaningful human-AI co-creative cultural studies require both advanced computational methods and philosophically informed safeguards, ensuring that the richness of human experience and the agency of communities remain central.

Keywords Cultural analytics · Cultural artifacts · Big data · Artificial intelligence · Co-creative cultural studies

Although they are both kingfishers, part of the same family, yet her beak is petite and elegant. Her eyes are sharp but sparkled with a hint of charm. Her delicate figure embodies the phrase “a swan’s grace,” yet she possesses incredible strength, easily lifting a three-ounce fish up to her branch. But most importantly, her ethereal attire earns her the title of “Dreamy Wings.” Her entire being is like an ethereal painting, a playful creation of nature using colors, lines, dots, strokes, and patches. One can only exclaim, “Absolutely enchanting. Is this real or an illusion?”

–In *The Philosophy of Awakening, Wild Wise Weird: The Kingfisher Story Collection*, Vuong (2025)–

Cultural Analytics by Lev Manovich (2020) outlines the recent developments and the historical roots of a new, exciting research field called cultural analytics. Cultural analytics emerges as a discipline that utilizes methods from computer

science, data visualization, and media arts for the exploration and analysis of cultural objects and their user interactions. In other words, cultural analytics studies cultures *at scale* via digital and computational methods. This field has gained traction in recent years (Black et al. 2024). The recent advancements and popular uses of generative AI have presented both opportunities and challenges for the studying of cultural events at scale with computational methods. For example, Bamman et al. (2024) explore how Large Language Models (LLMs) can be employed for classification tasks within cultural analytics, assessing their performance compared to traditional supervised methods. Their findings suggest that while LLMs are competitive for established tasks, they may underperform on novel ones, highlighting the need for critical evaluation of AI tools in cultural contexts. Meanwhile, Klein et al. (2025) admonishes against the overuse and the nuanced process of ‘elite capture’ of humanities scholarship by AI research, which manifests as the extraction of humanities methodologies and perspectives without corresponding investment in the disciplines that cultivate them. As Rettberg (2024) succinctly puts: “The current wave of AI is based on language, narratives, and culture; unchecked, this wave threatens to impoverish the world’s cultural narratives. “ Against this background, this essay reviews Manovich (2020)’s book then provide some critical discussions on how to harness the recent rise of AI while maintaining the humanistic spirit of cultural analytics.

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1 How humanities benefit from cultural analytics

Manovich argues the humanities can have twofold benefits from cultural analytics. First, the humanities, traditionally very qualitative, start seeing how the principles of the hard sciences can be applied to their research subjects with the rise of big data and accessible computational tools. Second, more importantly, as it inherits from methods and perspectives from a number of fields such as science and technology studies (STS), software studies, digital humanities, cultural analytics research not only apply these computational tools but also remain critical at questioning assumptions and biases laden in cultures and societies, including critically examining the “quantitative turn” of art and media studies as well. As the author stated: “Cultural analytics for me is the instrument to question all categorical boundaries.” (p.13). Indeed, as an emerging field, cultural analytics also probes theoretical and practical questions about the limitations of studying culture with computational methods and large-scale analysis. In 12 chapters, Manovich leads us through a series of key concepts, methods, seminal studies, theoretical and practical issues that form the foundation for cultural analytics. For the sake of pedagogical utility, the twelve chapters are logically divided into three parts: part I familiarizes readers with examples of the field; part II serves as a methodological crash course of conceptual operations, choices, and constraints in generating datasets from cultural artifacts; and part III provides nuanced discussions on data visualization.

2 Studying culture at scale

Critically, the book poses many deep questions for and guides readers to see *culture at scale*. Since our lives now are increasingly digitized, tracked, and mediatized (Ho and Vuong, 2024), quantitative large-scale studies of cultural behaviors, including discussing, reading, listening, viewing, etc., have become possible. As clearly articulated in the twelve research challenges, carefully deployed computational methods on well-defined research questions in cultural and media studies can yield not only interesting results but also potentially lead to new discoveries of *regularities* in cultural movements. In other words, cultural analytics allow researchers to investigate not only contemporary phenomenon but also deep into the past and make prediction for the future. The author goes on to state that the discoveries and tracking of cultural DNAs and their manifested phenotypes (i.e., cultural behaviors) are possible.

Here, by frequently referring to the classic works in the humanities such as the famous critique of the cultural industry by Adorno and Horkheimer, which states “Culture today is infecting everything with sameness,” Manovich asks how the computational media analytics of today affect the products created by the culture industry and the choices of the consumers. Connecting the past and contemporary cultural studies, Manovich creates an infectious exciting feeling about what discoveries might be in front of us.

3 Computational studies on past cultural artifacts

Computational studies on past cultural artifacts can indeed yield astonishing new ideas and insights and help us rethink scientific concepts as well. For example, codifying the appearances of traditional cultural values and symbols (e.g., of Buddhism, Confucianism, Taoism) in folktales (Vuong et al. 2018) and satirical cartoons (Ho et al. 2021) have enabled researchers to reach deeper appreciation of concepts that represent cultural mixing mechanisms such as cultural additivity, cultural hybridity, cultural syncretism, etc.. Or computationally studying the digitized editorials of newspapers from 1827 to 1865 reveals the importance of the invisible labor of women as well as the Black press in the Abolitionist movement (Soni et al. 2021). Another example is Candia et al. (2019) show the average attention received by cultural products decays following a universal biexponential function, and biographies stay in our collective memory the longest (20–30 years), while music the shortest (about 6 years). Such an attempt to understand the law of collective memory decay can only be possible by analyzing the patterns of millions of citations received by academic articles and online attentions (votes, play counts, mentions, etc.) received by hundreds of thousands of cultural products (movies, music records, biographies).

The readers also benefit greatly from Manovich’s utmost care and rigor in differentiating among different commonly used terminologies such as algorithms, analytics, software. For example, Manovich refrains from using the word algorithms in association with the studies of culture and explains the preferred terms are software and analytics for they do not presume the use traditional algorithms, which executes a finite sequence of steps, nor do they assume interpretability. Or in discussing how to represent cultural phenomenon as data, Manovich prefers the term feature over variable, as the former does not assume a particular statistical paradigm.

Future researchers across disciplines can also derive great insights by following informational entropy-based value capture mechanism to combine both the theoretical and practical gaps in cultural analytics being explained in the

book (Vuong and Nguyen 2024). An example is the lack of organized efforts to build any systematic theory of cultural sampling (i.e., how to define representative cultural samples, what methods to construct such a sample, what statistical estimates of what can be learned with it and with what certainty, etc.). Indeed, critical science and technology studies readers can, at the same time, appreciate the new affordances of computational techniques for cultural studies and remain critical and ask big-picture questions critically on ontological and epistemological issues related to the new scientific and technological progress (Sismondo 2011).

The book does well in providing a balance between the philosophical discussions regarding cultural research in the age of data abundance and the technical, practical details that involved in data treatment and methods selection (Patton 2004). For example, Manovich points out one can see how the rise of advanced computational methods and big data allows for a hermeneutical cycle between the positivist and the constructivist approach. For example, to analyze museum paintings, one can create a dataset based on existing, socially acknowledged labels (where meanings have already been constructed) or one can use feature extraction algorithms to extract low-level features (e.g., edges, shapes, colors, objects, etc.) of the paintings then feed them into cluster-discovering algorithms to discover whether computer-generated features and clusters produce categories that match the existing ones or not.

4 From cultural analytics to philosophy of culture in the age of AI

Lev Manovich's *Cultural Analytics* provokes a fundamental reevaluation of how culture is conceived, categorized, and interpreted in the era of *big data and powerful AI agents*. Manovich challenges us to consider whether the very ontology of culture shifts when cultural artifacts—images, films, audio, video, texts—are reduced to quantifiable features, and whether algorithmic classifications might supplant traditional, context-laden categories (Manovich 2020). Philosophically, this raises questions about the nature of human cognition: can we truly “think without categories,” or do machine-learned schemas simply instantiate new, data-driven taxonomies (Couldry and Hepp 2017)?

Critics warn that metadata extraction risks decontextualizing lived experience and flattening rich cultural meanings into sterile data points (Boyd and Crawford 2012), yet advocates argue that computational methods unveil large-scale patterns that elude conventional hermeneutics and democratize cultural insight. Epistemologically, the act of structuring messy cultural phenomena into datasets introduces inevitable biases—from selection and encoding to algorithm

design—prompting calls for transparency and reflexivity in data-driven scholarship (Boyd and Crawford 2012).

Ethically, the power to predict trends and trace “cultural DNA” ignites debates about autonomy, surveillance, and the reproduction of social inequalities: whose narratives are amplified, whose are marginalized, and how might quantitative approaches reinforce existing hegemonies (Candia et al. 2019; Liu et al. 2025)? Counterarguments highlight that computational analysis need not replace humanistic inquiry but can be integrated with qualitative methods to preserve contextual nuance and ensure diverse voices are heard (Couldry and Hepp, 2017). Ultimately, *Cultural Analytics* demands a hybrid framework that balances the expansive reach and power of *big data and AI* with the critical humanistically informed safeguards—ensuring that the richness of cultural experience and the agency of individuals remain central in the digital age (Nguyen and Ho, 2025; Pham and Ho, 2025).

5 Conclusion

In sum, *Cultural Analytics*, as articulated by Manovich, compels us to reconsider not only how cultural artifacts are measured but also how they are communicated and governed within a society increasingly mediated and driven by advanced AI agents. Throughout the book, Lev Manovich illustrates and emphasizes the importance, even necessity of big data for the emerging ‘hard’ science of cultural studies. However, there is an equal emphasis on the fact that cultural data must not only be big but also rigorous and clever. Manovich continuously admonishes future researchers to think hard about the challenges of how cultural phenomenon can be represented as data to avoid the reductivism trap, as he quotes Gitelman and Jackson: “our zeal for more and more data can become a faith in their neutrality and autonomy, their objectivity...” This view aligned well with the caution against decontextualization of big data by Boyd and Crawford (2012). Beyond, the mere technical challenges of how cultural data should be generated, it is critical, as Klein (2025) succinctly articulated, scholars and society at large need to be educated and aware of “humanities extraction” (p.14), which the commodification of humanities scholarship within the rapidly evolving landscape of artificial intelligence (AI). While the incorporation of humanistic thinking into AI development is a positive development, the authors caution against superficial engagement that fails to acknowledge the depth and complexity of humanities scholarship. Here, human-AI co-creative cultural studies require both advanced computational methods and philosophically informed safeguards, ensuring that the richness of human experience and the agency of communities remain central.

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Declarations

Conflicts of interests The authors declare no competing interests.

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